

Computer Lab: SQL Server-10

Part 3: Data Definition (cont.)

The main SQL data definition language statements are:

CREATE DATABASE		DROP DATABASE
CREATE DOMAIN	ALTER DOMAIN	DROP DOMAIN
CREATE TABLE	ALTER TABLE	DROP TABLE
CREATE VIEW		DROP VIEW

These statements are used to create, change, and destroy the structures that make up the conceptual schema.

SQL ALTER TABLE Statement

The **ALTER TABLE** statement is used to add, delete, or modify columns in an existing table.

The **ALTER TABLE** statement is also used to add and drop various constraints on an existing table.

ALTER TABLE - ADD Column

To add a column in a table, use the following syntax:

```
ALTER TABLE table_name
ADD column_name datatype;
```

The following SQL adds an "Email" column to the "Customers" table:

Example

```
ALTER TABLE Customers
ADD Email varchar(50);
```

ALTER TABLE - DROP COLUMN

To delete a column in a table, use the following syntax:

```
ALTER TABLE table_name
DROP COLUMN column_name;
```

Example 8.4 ALTER TABLE

Adding and Dropping Columns

The following statement will add a column named Address to the Students table:

ALTER TABLE Students
ADD COLUMN Address VARCHAR(20);

You can drop an existing column with the statement:

ALTER TABLE Students
DROP COLUMN Address;

Removing a Table (DROP TABLE)

We can remove a redundant table from the database using the DROP TABLE statement, which has the format:

DROP TABLE TableName [RESTRICT CASCADE]
--

For example, to remove the Departments table we use the command:

DROP TABLE Departments;

Note, however, that this command removes not only the named table, but also all the rows within it. To simply remove the rows from the table but retain the table structure, use the DELETE statement instead.

The DROP TABLE statement allows you to specify whether the DROP action is to be cascaded or not:

- **RESTRICT** The DROP operation is rejected if there are any other objects that depend for their existence upon the continued existence of the table to be dropped.
- **CASCADE** The DROP operation proceeds and SQL automatically drops all dependent objects (and objects dependent on these objects).

For example, to drop a table:

DROP TABLE Departments RESTRICT;
DROP TABLE Departments CASCADE;

- **RESTRICT:** drop operation fails if the table is referenced by view/constraint definitions
- **CASCADE:** drop operation removes referencing view/constraint definitions

The total effect of a DROP TABLE with CASCADE can be very extensive and should be carried out only with extreme caution.

One common use of DROP TABLE is to correct mistakes made when creating a table. If a table is created with an incorrect structure, DROP TABLE can be used to delete the newly created table and start again.

8.4 Views

SQL CREATE VIEW Statement

In SQL, a view is a virtual table based on the result-set of an SQL statement.

A view contains rows and columns, just like a real table. The fields in a view are fields from one or more real tables in the database.

You can add SQL statements and functions to a view and present the data as if the data were coming from one single table.

A view is created with the **CREATE VIEW** statement.

CREATE VIEW Syntax

```
CREATE VIEW view_name AS
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

Note: A view always shows up-to-date data! The database engine recreates the view, every time a user query it.

Example 8.6

Create a view so that the manager at branch B003 can see only the details of staff who work in his or her branch office.

A horizontal view restricts a user's access to selected rows of one or more tables.

```
CREATE VIEW Manager3Staff
AS SELECT *
FROM Staff
WHERE branchNo = 'B003';
```

This creates a view called Manager3Staff with the same column names as the Staff table but containing only those rows where the branch number is B003, we would get the result table shown in Table 8.2.

Table 8.2 Data for view Manager3Staff.

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000.00	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000.00	B003
SG5	Susan	Brand	Manager	F	3-Jun-40	24000.00	B003

Example 6.7

Create a view of the staff details at branch B003 that excludes salary information, so that only managers can access the salary details for staff who work at their branch.

A vertical view restricts a user's access to selected columns of one or more tables.

```
CREATE VIEW Staff3  
AS SELECT staffNo, fName, lName, position, sex  
FROM Staff  
WHERE branchNo = 'B003';
```

Note that we could rewrite this statement to use the Manager3Staff view instead of the Staff table, thus:

```
CREATE VIEW Staff3  
AS SELECT staffNo, fName, lName, position, sex  
FROM Manager3Staff;
```

Either way, this creates a view called **Staff3** with the same columns as the Staff table, but excluding the salary, DOB, and **branchNo** columns. If we list this view we would get the result table shown in Table 8.3.

Table 8.3 Data for view Staff3.

staffNo	fName	lName	position	sex
SG37	Ann	Beech	Assistant	F
SG14	David	Ford	Supervisor	M
SG5	Susan	Brand	Manager	F

SQL Dropping a View

A view is deleted with the **DROP VIEW** statement.

SQL DROP VIEW Syntax

```
DROP VIEW view_name;
```

DROP VIEW causes the definition of the view to be deleted from the database. For example, we could remove the **Manager3Staff** view using the statement:

```
DROP VIEW Manager3Staff;
```

If CASCADE is specified, DROP VIEW deletes all related dependent objects, in other words, all objects that reference the view. This means that DROP VIEW also deletes any views that are defined on the view being dropped.

If RESTRICT is specified and there are any other objects that depend for their existence on the continued existence of the view being dropped, the command is rejected. The default setting is RESTRICT.

